

Find Laplace transforms of:

1. $(\sin t - \cos t)^2$.

2. $\cos^3 2t$.

3. $\cos (at + b)$.

4. $\sin 2t \cos 3t$.

5. $e^{-at} \sinh bt$.

6. $\cosh at \sin at$.

7. $\sinh 3t \cos^2 t$

8. $f(t) = \begin{cases} 4, & 0 \leq t \leq 1 \\ 3, & t > 1 \end{cases}$

9. $f(t) = \begin{cases} \sin t, & 0 < t < \pi \\ 0, & t > \pi \end{cases}$

10. If $L\{f(t)\} = \frac{1}{s(s^2+1)}$, find $L\{e^{-t}f(2t)\}$

11. $te^{2t} \sin 3t$

12. $te^{-2t} \sin 4t$

13. $2^t + \frac{\cos 2t - \cos 3t}{t} + t \sin t$

14. Using LT Prove that $\int_0^\infty \frac{e^{-at} - e^{-bt}}{t} dt = \log \frac{b}{a}$

15. Evaluate $L\left(\int_0^t \frac{\sin t}{t} dt\right)$

16. Express $f(t) = \begin{cases} \cos t, & 0 < t < \pi \\ 1, & \pi < t < 2\pi \\ \sin t, & t > 2\pi \end{cases}$ in terms of unit step function and hence find its LT

17. Express $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4t, & 2 < t < 4 \\ 8, & t > 4 \end{cases}$ in terms of unit step function and hence find its LT.

18. Evaluate $L\{(1 + 2t - 3t^2 + 4t^3)H(t - 2)\}$

19. Find LT of the full-wave rectifier $f(t) = E \sin \omega t$, $0 < t < \frac{\pi}{\omega}$, having the period $\frac{\pi}{\omega}$.

20. Find the LT of the triangular wave of period $2a$ given by $f(t) = \begin{cases} t, & 0 < t < a \\ 2a - t, & a < t < 2a \end{cases}$

Answers for the above questions:

1. $\frac{1}{s} - \frac{2}{s^2 + 4}$

2. $\frac{1}{4} \left[\frac{s}{s^2 + 9} + \frac{3s}{s^2 + 1} \right]$

3. $\left[\frac{s}{s^2+a^2} \cos b - \sin b \frac{a}{s^2+a^2} \right]$
4. $\frac{1}{2} \left[\frac{5}{s^2+25} - \frac{1}{s^2+1} \right]$
5. $\left[\frac{b}{(s+a)^2-b^2} \right]$
6. $\frac{1}{2} \left[\frac{a}{(s^2+a^2)^2-4a^2s^2} \right]$
7. $\frac{1}{4} \left[\frac{s-3}{(s-3)^2+1} - \frac{s+3}{(s+3)^2+1} + \frac{6}{s^2-9} \right]$
8. $\frac{4}{s} - \frac{e^{-s}}{s}$
9. $\frac{1+e^{-\pi s}}{s^2+1}$
10. $\frac{4}{(s-1)((s-1)^2+4)}$
11. $\frac{6(s-2)}{(s-2)^2+9}$
12. $\frac{6(s+2)}{(s+2)^2+16}$
13. $\frac{1}{s-\ln 2} + \frac{1}{2} \ln \left(\frac{s^2+9}{s^2+4} \right) + \frac{2s}{(s^2+1)^2}$
14. $\ln \left(\frac{b}{a} \right)$
15. $\frac{\frac{\pi}{2} - \tan^{-1}s}{s}$
16. $\frac{s(1+e^{-\pi s})+e^{-2\pi s}}{s^2+1} + \frac{e^{-\pi s}-e^{-2\pi s}}{s}$
17. $\frac{2}{s^3} + e^{-2s} \left[\frac{4}{s} - \frac{2}{s^3} \right] - e^{-4s} \left[\frac{4}{s^2} - \frac{8}{s} \right]$
18. $e^{-2s} \left[\frac{24}{s^4} + \frac{42}{s^3} + \frac{38}{s^2} + \frac{25}{s} \right]$
19. $\frac{E\omega}{s^2+\omega^2} \left[\frac{1+e^{-\pi s/\omega}}{1-e^{-\pi s/\omega}} \right]$
20. $\frac{1}{s^2} \left[\frac{1-e^{-as}}{1+e^{-as}} \right]$